7. A lab assistant prepared a solution by adding a calculated quantity of HCl gas at 25°C to get a solution with $[H_3O^+] = 4 \times 10^{-5} M$. Is the solution neutral (or) acidic (or) basic.
8. Calculate the pH of 0.04 M HNO_3 Solution.
9. Define solubility product
10. Define ionic product of water. Give its value at room temperature.
11. Explain common ion effect with an example
12. Derive an expression for Ostwald's dilution law
13. Define pH
14. Calculate the pH of $1.5 \times 10^{-3} M$ solution of $Ba(OH)_2$
15. 50ml of 0.05M HNO_3 is added to 50ml of 0.025M KOH . Calculate the pH of the resultant solution.
16. The K_a value for HCN is 10^{-9} . What is the pH of 0.4M HCN solution?
17. Calculate the extent of hydrolysis and the pH of 0.1 M ammonium acetate Given that $K_a = K_b = 1.8 \times 10^{-5}$
18. Derive an expression for the hydrolysis constant and degree of hydrolysis of salt of strong acid and weak base
19. Solubility product of Ag_2CrO_4 is 1×10^{-12} . What is the solubility of Ag_2CrO_4 in 0.01M $AgNO_3$ solution?
20. Write the expression for the solubility product of $Ca_3(PO_4)_2$
21. A saturated solution, prepared by dissolving $CaF_2(s)$ in water, has $[Ca^{2+}] = 3.3 \times 10^{-4} M$ What is the K_{sp} of CaF_2 ?
22. K_{sp} of AgCl is 1.8×10^{-10} . Calculate molar solubility in 1 M $AgNO_3$
23. A particular saturated solution of silver chromate Ag_2CrO_4 has $[Ag^+] = 5 \times 10^{-5}$ and $[CrO_4]^{2-} = 4.4 \times 10^{-4} M$. What is the value of K_{sp} for Ag_2CrO_4 ?
24. Write the expression for the solubility product of Hg_2Cl_2.
25. K_{sp} of Ag_2CrO_4 is 1.1×10^{-12}. what is solubility of Ag_2CrO_4 in 0.1M K_2CrO_4 .
26. Will a precipitate be formed when 0.150 L of 0.1M $Pb(NO_3)_2$ and 0.100L of 0.2 M NaCl are mixed? $K_{sp}(PbCl_2) = 1.2 \times 10^{-5}$.
27. K_{sp} of $Al(OH)_3$ is $1 \times 10^{-15} M$. At what pH does $1.0 \times 10^{-3} M$ Al^{3+} precipitate on the addition of buffer of NH_4Cl and NH_4OH solution?

XII U8-Ionic equilibrium.indd 31 16-12-2022 19:34:46